

Table 64. USART (SPI mode) characteristics

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
f _{CK}	USART clock frequency	Master mode	-	-	6.0	MHz
		Slave receiver mode	-	-	16.0	
		Slave transmitter	-	-	16.0	
t _{su(NSS)}	NSS setup time	Slave mode	T _{ker} ⁽¹⁾ + 1	-	-	ns
t _{h(NSS)}	NSS hold time	Slave mode	2	-	-	
t _{w(CKH)}	CK high time	Master mode	1 / f _{CK} / 2 - 1	1 / f _{CK} / 2	1 / f _{CK} / 2 + 1	
t _{w(CKL)}	CK low time					
t _{su(MI)}	Data input setup time	Master mode	16	-	-	
t _{su(SI)}		Slave mode	1.5	-	-	
t _{h(MI)}	Data input hold time	Master mode	0	-	-	
t _{h(SI)}		Slave mode	0	-	-	
t _{v(SO)}	Data output valid time	Slave mode 2.7 V < VDD < 3.6 V	-	12.0	19	
		Slave mode 2.0 V < VDD < 3.6 V	-	12.0	13	
t _{v(MO)}		Master mode	-	2.0	4	
t _{h(SO)}	Data output hold time	Slave mode	9.5	-	-	
t _{h(SO)}		Master mode	0.5	-	-	

1. T_{ker} is the `usart_ker_ck_pres` clock period

Figure 29. USART timing diagram in SPI master mode

